

PowerScope

POWER-SYSTEM SIMULATION LAB · VERSION 1.2.1

Interactive platform for power-system simulation,
protection studies and hardware-in-the-loop teaching

README · User guide

K. Kumar, A. Lekić and N.-K. C. Nair
The University of Auckland, New Zealand · Delft University of Technology (TU Delft), the Netherlands



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1. About PowerScope

PowerScope is an interactive power-system simulation lab for research, teaching and hardware-in-the-loop experiments. It solves power flow, short circuits, RMS and EMT dynamics, protection, IEC 61850 digital substations, power quality, DC grids and extreme-event resilience on benchmark and user-built networks. Results appear on an animated single-line diagram, in publication-quality figures and tables, and in a plain-language panel that explains what happened and why.

The whole application is **one self-contained HTML file**. It needs no installation and no server: open it in a modern browser and every engine runs locally on your computer.

At a glance

Item	Details
Benchmark networks	11, from a single machine against an infinite bus to the IEEE 39-bus system, the CIGRE MV microgrid, an IEC 61850 substation, a four-terminal HVDC grid, a reduced New Zealand grid and your own networks
Studies	59 studies in 13 categories (section 5)
Simulation domains	Static (phasor), RMS (electromechanical dynamics), EMT (instantaneous waveforms), Auto
Engines	Built-in solvers validated against MATPOWER and analytic references; time-domain EMT engine; optional pandapower in the browser; script export for pandapower, PyPSA and VeraGrid
Frequency	50 Hz or 60 Hz for every network
Extra views	Power-plant operation (steam and hydro units), hardware-in-the-loop panel, system check, network editor
Figures	PNG or TIFF at 150, 300 or 600 DPI, sized for IEEE columns or slides
Version	1.2.1

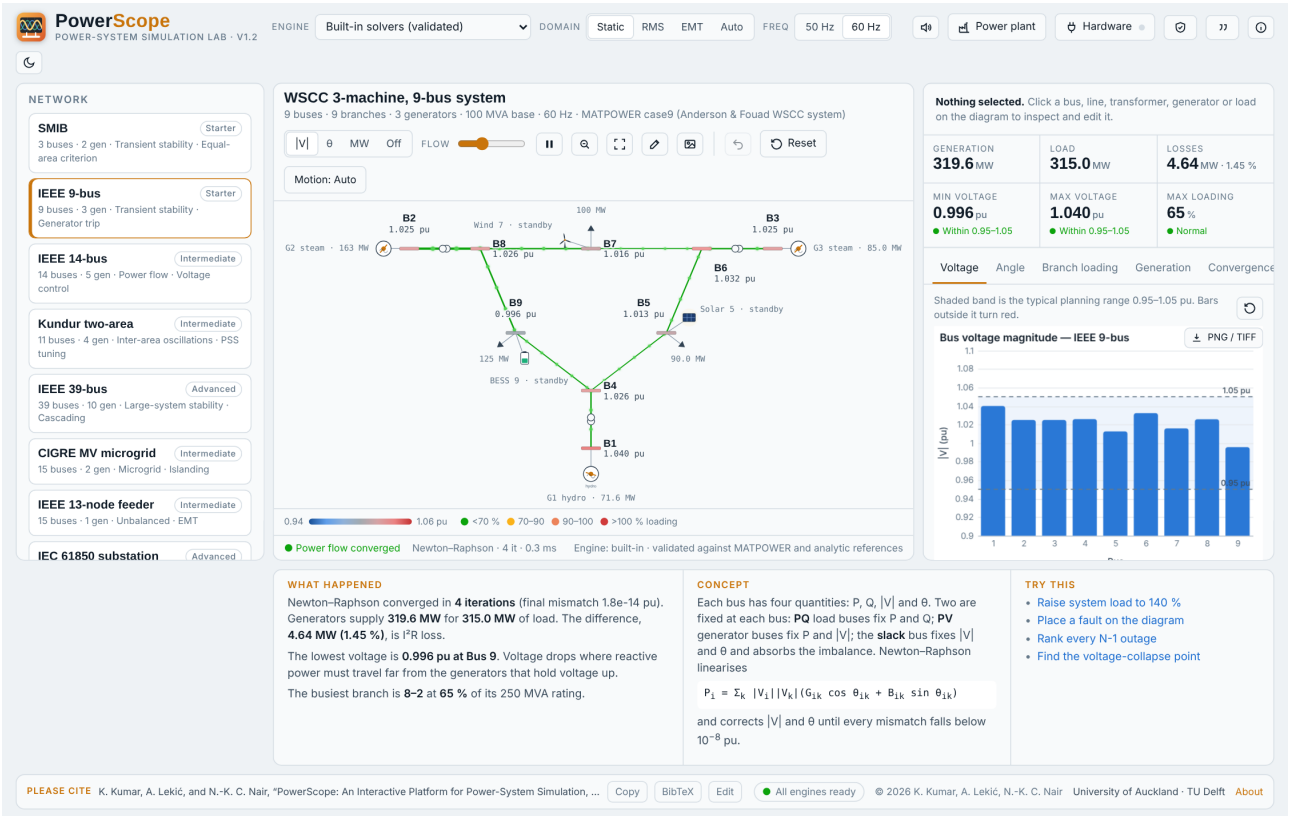


Figure 1. The main window: network list (left), animated single-line diagram with timeline (centre), key results and tabs (right), explanation panel (bottom) and citation bar.

2. Getting started

Requirements

- A current desktop browser. **Chrome or Edge** are recommended: they also support the USB serial link for the Arduino panel.
- No installation and no internet connection for the built-in engines. An internet connection is needed only once, for the optional pandapower engine (about 40 MB, downloaded on first use).
- A screen of at least 1366 × 768. The layout also adapts to smaller screens.

Opening the file

Double-click `PowerScope_v1.2.1.html` or drag it into a browser window. The page opens on the IEEE 9-bus system with an AC power flow already solved. Use the **System check** button (shield icon, top right) to confirm every engine loaded; it should report *Everything required works*.

Your first study in five steps

- **Pick a network** in the left-hand list (for example IEEE 39-bus).
- **Pick a study** from the study catalogue below the network list (for example *Stability → Transient stability*).
- **Set it up** in the settings panel: fault location, type, clearing time, models and so on. Clicking a bus, line or generator on the diagram usually selects it as the event location.
- **Watch the result.** The study runs automatically. Press play on the timeline to animate the event; the diagram, charts and key results move together.
- **Read the explanation** at the bottom (*What happened · Concept · Try this*). Each *Try this* link changes one setting and reruns, so you can explore cause and effect quickly.

Undo and reset

Undo (Ctrl+Z) reverses the last change. **Reset** returns the network and the study to their default case, removing faults, outages, edits and setting changes. Each study also has a button that puts only its own settings back to defaults.

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3. The interface

Top bar

Control	What it does
Engine	**Built-in solvers (validated)**, **EMT engine (time domain)** or **pandapower (Python cross-check)**.
Domain	**Static** phasor studies, **RMS** phasor dynamics (electromechanical), **EMT** instantaneous waveforms, or **Auto**, which picks the right domain for the chosen study.
Freq	Runs the network at 50 Hz or 60 Hz.
Sound	Operational sounds for breakers, isolators and faults; click to mute.
Power plant	Opens the power-plant operation view (section 7).
Hardware	Opens the hardware-in-the-loop window (section 8). The dot shows the link status.
Shield icon	**System check**: verifies the engines against reference results and lists browser features (section 10).
Quote icon	**How to cite PowerScope**: citation text and BibTeX (section 12).
i icon	**About PowerScope**: authors, institutions, citation, licence and benchmark-data sources.
Moon icon	Switches between the light and dark themes.

Diagram toolbar

Control	What it does
|V| · θ · MW · Off	Chooses the labels on the diagram: voltage magnitude, voltage angle, power flows, or none.
Flow slider	Speed of the animated power-flow particles.
Pause	Pauses the animation.
Fit / Full screen	Fits the diagram to the view, or fills the screen with it.
Pencil	Edits the network: add, move or delete buses, lines, transformers, generators and loads.
Picture	Exports the diagram as PNG or TIFF.
Undo / Reset	Undo the last change (Ctrl+Z) / reset everything to the default case.
Motion	**Auto** follows the computer's reduce-motion setting; **On** always animates; **Off** stops movement.

Reading the diagram

- Bus colours follow voltage (blue low, red high; the scale appears under the diagram). During faults and sags the colour shows retained voltage.
- Line colours follow loading: green below 70 %, amber 70–90 %, orange 90–100 %, red above 100 %.
- Breakers drawn filled are closed, hollow are open. Tripped relays and open breakers appear on the diagram and in the timeline strip.
- Click any element to inspect and edit it in the right-hand panel; changes rerun the current study.

Timeline and results

Event studies show a timeline under the diagram, split into stages (pre-fault, fault, after clearing…). Drag it, press play, or choose a speed. Charts marked with the timeline follow it. The right-hand panel shows key results with a traffic-light verdict, then tabs of charts and tables. Every chart zooms by dragging (double-click to reset) and downloads with its **PNG / TIFF** button; tables download as CSV.

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4. Networks

Choose a network in the left-hand list. Benchmark data are used for verification; the IEC 61850 substation, the HVDC grid and the New Zealand grid are illustrative PowerScope models with typical data. **The New Zealand network is not Transpower data.**

Network	Size	Description	Source
Single machine – infinite bus	3 buses 3 branches 2 gen 60 Hz	One 2220 MVA generator feeding a strong grid through a step-up transformer and two parallel lines. The smallest system that shows synchronism, power-angle curves and critical clearing time.	Kundur, Power System Stability and Control, Example 13.2
WSCC 3-machine, 9-bus system	9 buses 9 branches 3 gen 60 Hz	Three generators (one hydro, two steam) supplying three loads over a 230 kV ring. The standard first case for multi-machine stability.	MATPOWER case9 (Anderson & Fouad WSCC system)
IEEE 14-bus test system	14 buses 20 branches 5 gen 60 Hz	A portion of the American Electric Power system (1962): 132 kV and 33 kV networks, two generators and three synchronous condensers holding voltage.	MATPOWER case14 (University of Washington archive)
Kundur two-area, four-machine system	11 buses 12 branches 4 gen 60 Hz	Two identical areas, each with two 900 MVA units, joined by a weak 220 km tie. About 400 MW flows from Area 1 to Area 2 and excites a slow inter-area mode.	Kundur, Power System Stability and Control, Example 12.6
IEEE 39-bus New England system	39 buses 46 branches 10 gen 60 Hz	A reduced model of the New England 345 kV grid: 10 generators (bus 39 represents the interconnection to New York) and 19 loads totalling 6.1 GW.	MATPOWER case39 (Athay et al., 1979)
CIGRE European MV benchmark with DER	15 buses 17 branches 2 gen 50 Hz	A 20 kV distribution network fed from 110 kV through two 25 MVA transformers: an urban cable feeder with PV, a 1.5 MW wind turbine, fuel cells, a CHP diesel unit and two batteries, plus a rural overhead feeder. Switches S1–S3 (normally open) can close loops. Buses 7–11 form a natural microgrid.	CIGRE TB 575 (2014) MV benchmark; data as pandapower create_cigre_network_mv(with_der='all')
IEEE 13-node test feeder (unbalanced, 4.16 kV)	15 buses 14 branches 1 gen 60 Hz	A short, heavily loaded 4.16 kV feeder with single- and two-phase laterals, unbalanced spot and distributed loads, a voltage regulator, an in-line 4.16/0.48 kV transformer and shunt capacitors. The three-phase data are exact; single-line studies use a balanced equivalent.	IEEE PES Distribution Test Feeders (Kersting, 2001); published power-flow solution
Northgate 220/33 kV digital substation	10 buses 12 branches 2 gen 50 Hz	A 220/33 kV substation with a full IEC 61850 process bus: four 220 kV line bays and two transformer bays, each with a merging unit, a switchgear controller, protection and a bay controller. The surrounding grid gives the protection something real to protect.	PowerScope illustrative model · typical 220 kV line and 33 kV feeder data
Four-terminal ±320 kV MMC grid in a 400 kV system	5 buses 7 branches 2 gen 50 Hz	Five 400 kV regions joined by long AC lines, with a meshed ±320 kV DC grid of four modular multilevel converters (two cables, three overhead lines) carrying bulk power in parallel with the AC paths.	PowerScope illustrative model · typical 400 kV OHL and ±320 kV cable / OHL data

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Network	Size	Description	Source
New Zealand 220 kV backbone with the HVDC Inter-Island link	16 buses 42 branches 8 gen 50 Hz	Seventeen 220 kV regions of the North and South Islands. The islands are separate AC systems with their own frequency, joined only by the Benmore–Haywards HVDC link, which usually sends South Island hydro north.	PowerScope illustrative model of the NZ grid · approximate geography, loads and generation (not Transpower data)
Your network (editor)	2 buses 1 branches 1 gen 50 Hz	Start from a grid in-feed and a load, then add buses, lines, transformers, generators and loads on the diagram. Every study runs on what you build; Save keeps it as a file.	built in PowerScope

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5. Study catalogue

All 59 studies run on any suitable network. Each one has its own settings, key results, chart tabs and explanation.

Power flow & network

Study	What you can explore
AC power flow (Newton–Raphson)	Voltages, flows, losses and loading; generator Q limits and load scaling.
DC power flow	Linearised flows compared with the AC solution, with the error on every branch.
N-1 contingency ranking	Every single outage ranked by overloads and voltage violations.
PV / QV curves · voltage collapse	Continuation power flow to the nose point; reactive margin at a bus.
Tap-changer & reactive control	On-load tap changers, capacitor banks and voltage control.
Switching & operator error	Switching sequences, including mistakes and their consequences.
pandapower cross-check	Runs real pandapower (Python, in the browser) on the same case and compares results.
Unbalanced three-phase power flow	Phase-by-phase solution, voltage unbalance (VUF) and phasors; IEEE 13-node feeder against the published solution.

Electrical faults

Study	What you can explore
Shunt faults LG · LL · LLG · LLL · LLLG	Phase-domain fault currents and voltages, with fault and earthing resistance.
IEC 60909 / ANSI short-circuit levels	Ik″, ip, Ib, Ith and ANSI duties at every bus.
Open-conductor & broken-conductor	Open phases, with or without a fault on the broken end; negative sequence.
Cross-country & evolving faults	Simultaneous faults at two places and faults that change type.
EMT faults · arcing · high-impedance	Instantaneous waveforms: DC offset, arc models, high-impedance faults and their harmonics.
Intermittent / restriking faults	Restriking earth faults under isolated, resonant, resistance and solid earthing.

Stability

Study	What you can explore
Transient stability · CCT · equal-area	Rotor angles after a fault, critical clearing time, equal-area view on the SMIB.
Small-signal · eigenvalues · modes	Eigenvalues, damping, mode shapes and participation factors.
Frequency stability · inertia · RoCoF	Frequency nadir, RoCoF and settling after a generation loss.
Inter-area & local oscillations · PSS	Inter-area modes and power-system stabiliser tuning.
Voltage stability & weak grids (dynamic)	Dynamic voltage recovery and collapse in weak systems.

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Renewables & variability

Study	What you can explore
Solar PV day · clouds · seasons	Daily PV profiles with cloud transients and seasonal effects.
Wind · turbulence · gusts · cut-out	Wind output with turbulence, gusts and storm cut-out.
Hydro inflow · dry/wet year · spill	Water value, storage and spill over dry and wet years.
Battery storage · FFR · smoothing	Fast frequency response, smoothing and arbitrage.
Renewable penetration 0–100 %	System behaviour as renewables replace synchronous generation.
Grid-forming vs grid-following	Inverter control in strong and weak grids; system strength.

Islanding & restoration

Study	What you can explore
Generator trip · N-1 / N-2 · AUFLS	Generator loss with governor response and under-frequency load shedding.
Islanding · anti-islanding · resync	Island formation, anti-islanding detection and resynchronisation.
Black-start test & restoration path	Automatic restoration plan from a black-start unit, checked dynamically.

Equipment faults

Study	What you can explore
Busbar, cable, insulator & conductor-clash faults	Equipment-specific fault scenarios.
Transformer energisation & inrush	Inrush current, residual flux, controlled closing and second-harmonic blocking.
Generator: stator, rotor, loss of field, reverse power	Generator internal faults and abnormal conditions.
Breaker failure · stuck pole · restrike	Breaker-failure protection, stuck poles and restrikes.
CT saturation · VT fuse fail · ferroresonance	Instrument-transformer problems and their effect on protection.

Protection

Study	What you can explore
Overcurrent · IEC/IEEE curves · TCC	Inverse-time curves and time–current coordination charts.
Distance protection · R-X zones	Mho and quadrilateral zones, fault resistance, infeed and load encroachment.
Differential protection	Line, transformer and busbar differential with bias characteristics.
Coordination & selectivity	Automatic grading and the full trip sequence for any fault.
Maloperation · power swing · cascading	Power swings, load encroachment and cascading trips.

Digital substation (IEC 61850)

Study	What you can explore
GOOSE messaging · trip transfer time	GOOSE retransmission and end-to-end trip time.
Sampled Values · merging units	9-2LE streams, merging units and network load.
PTP time sync · holdover	Precision time protocol, loss of sync and holdover.
Comms fault injection · PRP/HSR	Network failures and redundancy protocols.
Virtual FAT / SAT test	Factory and site acceptance testing of the protection scheme.
Spoofed GOOSE / SV injection	Cyber-attack scenarios on the process bus and their detection.

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Power quality

Study	What you can explore
Sags · swells · transient overvoltage (EMT)	Half-cycle RMS, ITIC curve and sag types; capacitor switching.
Harmonics · THD · resonance scan	Harmonic flows, THD and frequency scans for resonance.
Flicker & unbalance	IEC flickermeter Pst and voltage unbalance.

DC & HVDC

Study	What you can explore
Pole-pole / pole-ground faults · di/dt	DC fault currents, rate of rise and converter blocking.
DC breaker interruption	Hybrid and mechanical DC breakers clearing a DC fault.
LCC commutation failure	Line-commutated converter commutation failure after AC faults.

Operations & control

Study	What you can explore
State estimation & bad data	Weighted least squares, observability and bad-data detection.
Optimal dispatch · unit commitment · SCOPF	Economic dispatch, commitment, security constraints and nodal prices.
AGC & reserves	Automatic generation control, tie-line bias and reserves.
SCADA / comms loss	Operation with lost telemetry and control.

Validation

Study	What you can explore
Validation report (all engines)	Runs every engine against its references and reports the results.

Environment & New Zealand

Study	What you can explore
Lightning strike & back-flashover	Travelling waves, insulator flashover and surge arresters.
Storm · earthquake multi-contingency	Multiple simultaneous outages from extreme events.
Vegetation fault on resonant earthing	High-impedance vegetation faults on compensated networks.
NZ scenarios: HVDC pole trip, Tiwai, AUFLS	Inter-island HVDC events, Tiwai Point smelter trip and AUFLS.

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6. Engines and simulation domains

Engine / domain	Use it for	Notes
Built-in static solvers	Power flow (Newton–Raphson, DC), short circuit (IEC 60909, sequence networks, phase domain), state estimation, optimisation	Validated against MATPOWER and analytic references
RMS dynamics	Stability, frequency, oscillations, islanding, black start, inverter control	Phasor (electromechanical) time domain; synchronous machines, exciters, governors, PSS, inverters
EMT engine	Arcing and restriking faults, inrush, sags and switching, lightning, DC faults	Instantaneous values, microsecond time steps; trapezoidal companion models
pandapower	Independent cross-check of power flow and short circuit	Real Python running in the browser (Pyodide); downloads about 40 MB on first use
Script export	Rerun the same case in pandapower, PyPSA or VeraGrid	Download buttons in the settings of the relevant studies

Choosing the domain

Auto is the recommended setting: each study opens in the domain it needs. Choose **Static**, **RMS** or **EMT** to browse the studies that use that domain. RMS results average out waveforms; EMT shows DC offsets, travelling waves, arcs and harmonics, but takes more computing time for large networks.

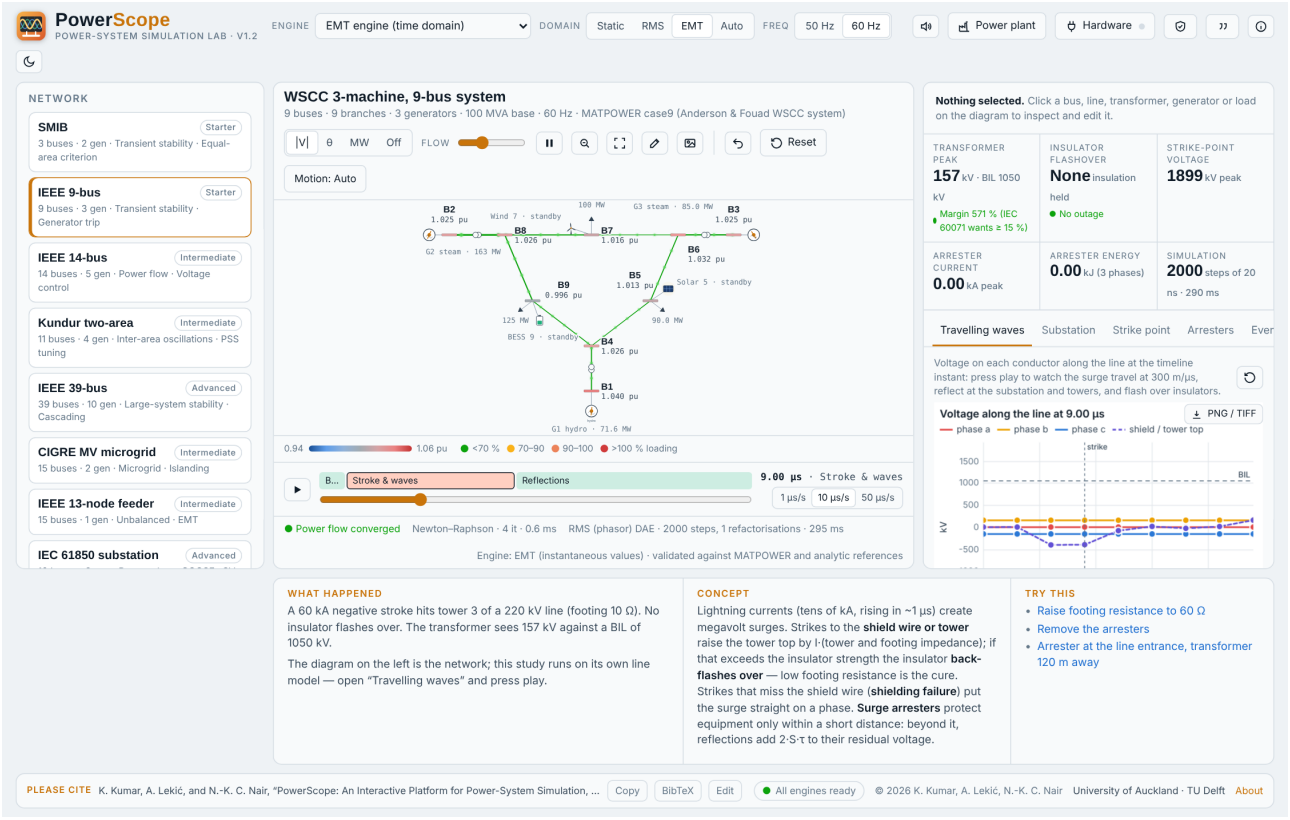


Figure 2. EMT study: a lightning strike on a 220 kV line, with the travelling-wave view and insulation margins.

7. Power-plant operation

The **Power plant** button opens a process and electrical model of a generating unit, from cold shutdown to full load. Choose a **steam unit** (coal or gas, 250 MW, 18 kV, 3000 rpm) or a **hydro unit**, and run time at 1×, 5×, 20× or 60×.

Procedure

- **1 Start auxiliaries** — cooling-water, condensate and boiler-feed pumps and draught fans from the station-service transformer.
- **2 Light the boiler**, **3 latch and roll the turbine**, **4 excite the generator** (close field breaker 41).
- **5 Synchronise** — match speed and voltage on the synchroscope and close generator breaker 52G, supervised by sync check 25. An auto-synchroniser can do this for you.
- **6 Load the unit** with the load set-point and ramp rate, **7 normal operation**, **8 shut down**.
- **Quick start** skips to the unit synchronised at 60 % load; **Reset plant** returns to cold shutdown.

Protection and events

Generator protection functions 12 (overspeed), 25 (sync check), 32 (reverse power), 40 (loss of field), 51V, 78 (out of step), 81 (frequency), 87G (differential) and 86 (lockout) are shown on the single-line diagram; click any of them for details. The events panel applies grid frequency and voltage steps, line faults with a chosen clearing time, load rejection, turbine trip, loss of field and stator winding faults. A training option bypasses sync check 25 to show what out-of-phase closing does.

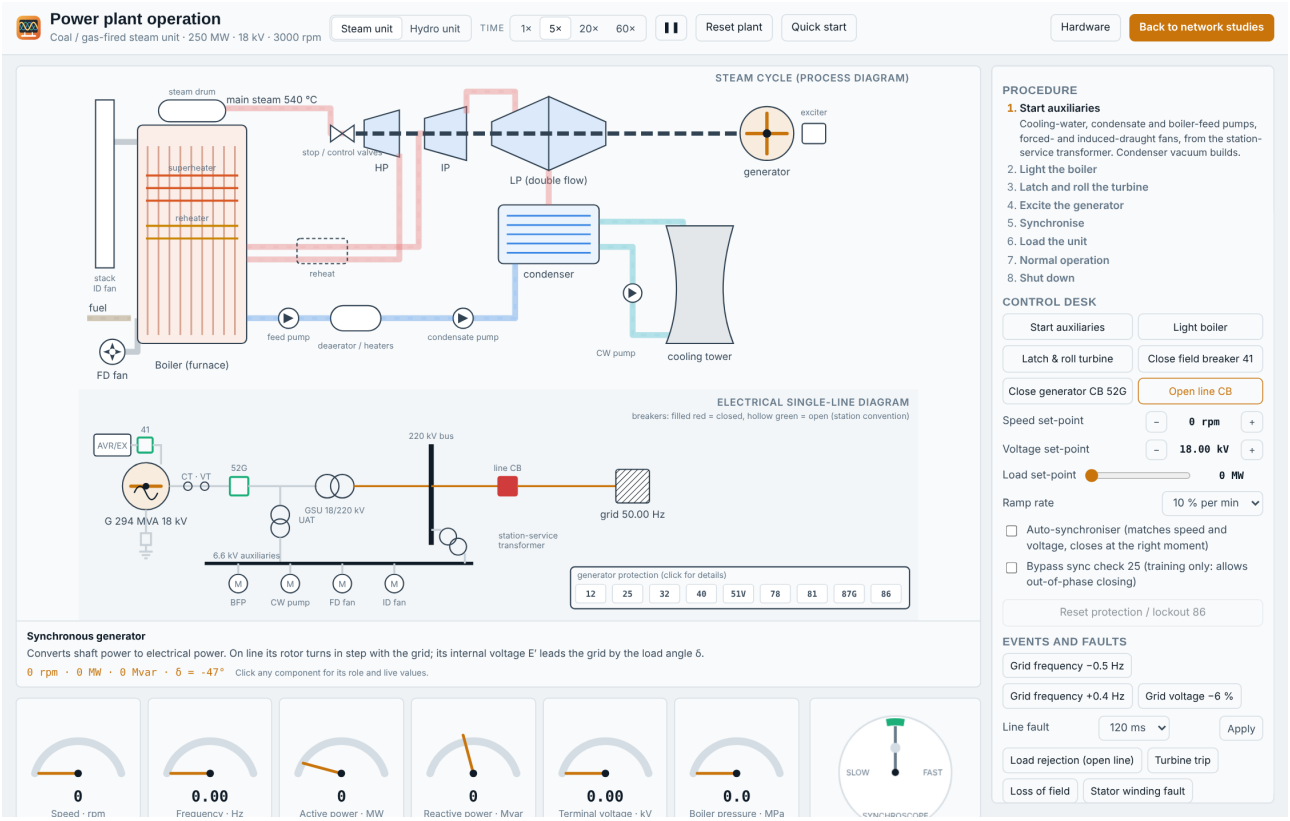


Figure 3. Power-plant view: steam cycle, electrical single-line diagram, procedure, control desk and gauges.

8. Hardware-in-the-loop

The **Hardware** window connects a physical panel of LEDs, buttons, a knob and a meter to PowerScope. LEDs follow breaker states, buttons operate breakers, the knob sets load (or the MW set-point in the power-plant view) and the meter shows a voltage or MW. **Download HIL kit (.zip)** gives the Arduino sketch, the Raspberry Pi bridge and a README.

Connection	How
Arduino (USB)	Upload `powerscope_hil.ino` with the Arduino IDE 2. In Chrome or Edge choose Hardware → Arduino (USB) → Connect, pick the port, then Test (LEDs blink three times, the meter sweeps). The port is remembered.
Raspberry Pi / bridge	Run `install.sh` (Windows: install.bat), then `pi_bridge.py --gpio` or `--serial`. Connect to `ws://raspberrypi.local:8765`.
Simulated panel	A virtual breaker panel that speaks the same protocol; no hardware needed.

Default Arduino wiring

Function	Pins
Breaker LEDs (branches 1–4)	D2–D5, each through 330 Ω to GND
Push-buttons (toggle branches 1–3)	D6–D8 to GND
Fault button (hold 1.5 s to reset)	D12 to GND
Load / MW knob	10 kΩ potentiometer on A0 (ends to 5 V and GND); sets load 50–150 %
Meter	D9 (PWM): |V| of the first bus, 0.90–1.10 pu full scale
Heartbeat	D13 (on-board LED) blinks on every telemetry frame

Protocol

One message per line, as text or JSON, at 115 200 baud by default. The window's Setup, Mapping, Wiring & links and Diagnostics tabs show each step, which panel point operates which branch, and the live traffic.

```
PowerScope -> device
  HELLO PowerScope 1.1 net=ieee9 buses=9 branches=9 gens=3
  T t=0.000 f=60.000 V=1.040,1.025,... B=1,1,0,... G=1,1,1 L=100 P=1,1,0,1 M=0.52
  TEST | PING | EV OPEN 3 | EV CLOSE 3 | EV TRIP G2 | EV FAULT 7 | EV CLEAR | EV RESET

device -> PowerScope
  OPEN n | CLOSE n | TOGGLE n | TRIP Gn | RESTORE Gn | LOAD pct | FAULT bus
  CLEAR | RESET | SYNC | PING | PONG | INFO ... | NAME=value (e.g. A0=512)
```

Safety. The kit is for low-voltage teaching hardware only. Never connect PowerScope outputs to real protection or switchgear control circuits.

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9. Exporting results and building networks

Figures

Every chart and the diagram export through the same dialog. The figure is redrawn at the target resolution, not upscaled.

Option	Choices
Format	PNG (with embedded DPI) or TIFF
Resolution	150, 300 or 600 DPI
Size	IEEE single column (3.5 in), IEEE double column (7.16 in), slide 16:9 (10 × 5.6 in), or custom
Style	Paper style or current theme; optional serif font (Times); include or omit the title

Data

Tables download as CSV. The pandapower study can also download an offline copy of PowerScope and a pandapower script; dispatch studies download a PyPSA script, and dynamic studies a VeraGrid script, so the same case can be rerun in those tools.

Custom builder

Select **Custom builder** in the network list, or press the pencil on any diagram. The editor palette adds a **bus**, **line**, **transformer**, **generator** or **load**; **Move** and **Delete** rearrange the network. Click empty space to place a bus, then click elements to edit their data in the right-hand panel. Every study runs on what you build, and **Save** keeps it as a file you can open again later.

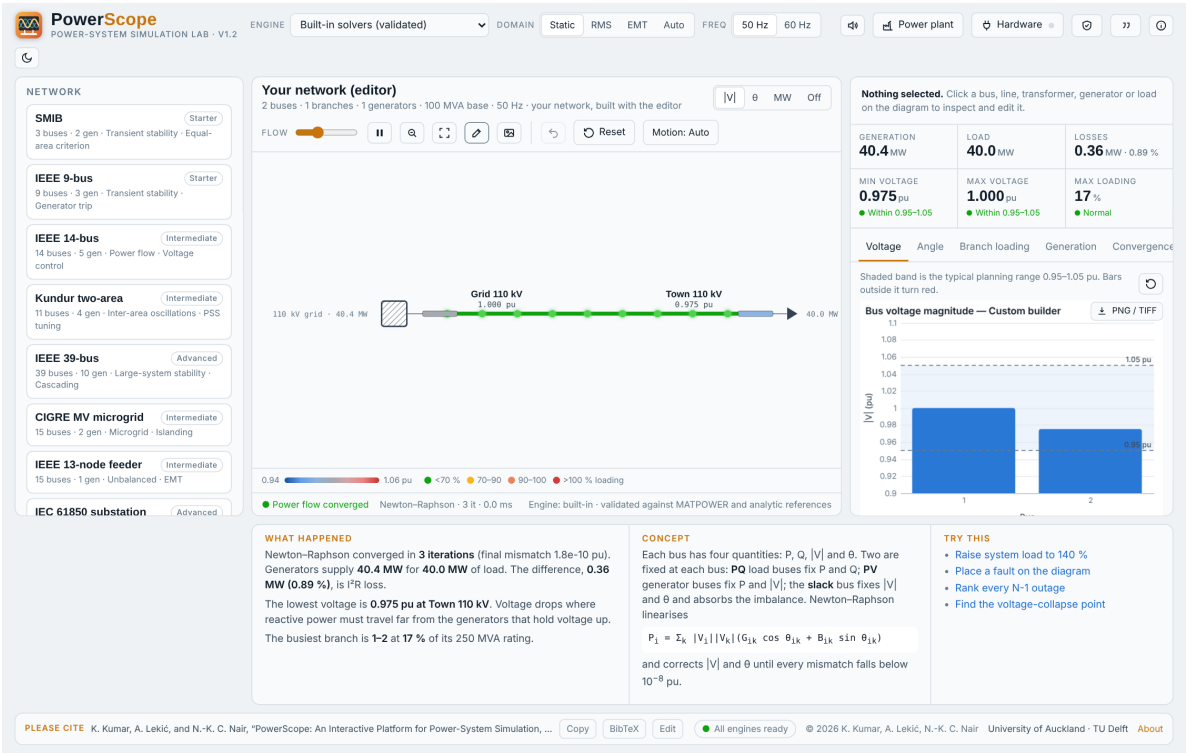


Figure 4. The Custom builder network. The pencil button switches the diagram into the editor.

10. Verification

System check (shield icon) confirms that every engine is loaded and reproduces reference results, then lists the browser features available. The *Validation report* study runs the full set of engine comparisons. The results below are from version 1.2.1 in Chrome.

Check	Expected	PowerScope
Power flow, IEEE 9-bus: slack generation	71.64 MW	71.641 MW
Power flow, IEEE 9-bus: total losses	4.641 MW	4.641 MW
Power flow, IEEE 14-bus: largest |V| difference	0 pu	0.0013 pu
Power flow, IEEE 39-bus: total losses	43.641 MW	43.641 MW
Power flow, IEEE 39-bus: slack unit (bus 31)	677.87 MW	677.87 MW
Short circuit: peak factor κ at R/X = 0.1	1.746	1.746
Short circuit, IEEE 14-bus bus 4: Ik3 vs Thevenin impedance	5.6476 kA	5.6476 kA
Short circuit, IEEE 9-bus bus 5: phase domain vs IEC 60909	2.1202 kA	2.1202 kA
Protection: IEC standard inverse, M = 10, TMS = 1	2.971 s	2.9706 s
Protection: IEC very inverse, M = 10, TMS = 1	1.5 s	1.5 s
Protection: IEEE very inverse, M = 5, TD = 1	1.308 s	1.3081 s
Power quality: flickermeter Pst, IEC test points (worst of 4)	1	1.0202
Power quality: capacitor-bank parallel resonance (IEEE 14-bus, bus 9)	3.80 order	3.94 order
Power quality: unbalance of a 10 MVA phase-to-phase load	1.4126 %	1.4126 %
DC: four-terminal grid, converter balance = line losses	8.8223 MW	8.8223 MW
DC: LCC inverter extinction angle equals its setting	18°	18°
State estimation: WLS with exact measurements (largest |V| error)	0 pu	2.2 × 10□¹⁵ pu
IEC 61850: GOOSE repetition 0, T1 … 16T1 (T1 = 2 ms)	32 ms	32 ms
IEC 61850: 9-2LE stream at 50 Hz, 4000 frames/s of 146 bytes	4.672 Mb/s	4.672 Mb/s
NZ pack: frequency after a 30 MW loss (load relief + droop)	49.571 Hz	49.571 Hz
EMT: RL source switched onto a resistor, steady-state peak	1116.4 V	1116.4 V

Small differences (IEEE 14-bus voltages, resonance order, flicker) come from published data rounding, model detail and the tolerances of the reference methods; all are within each check's pass limit.

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11. Gallery

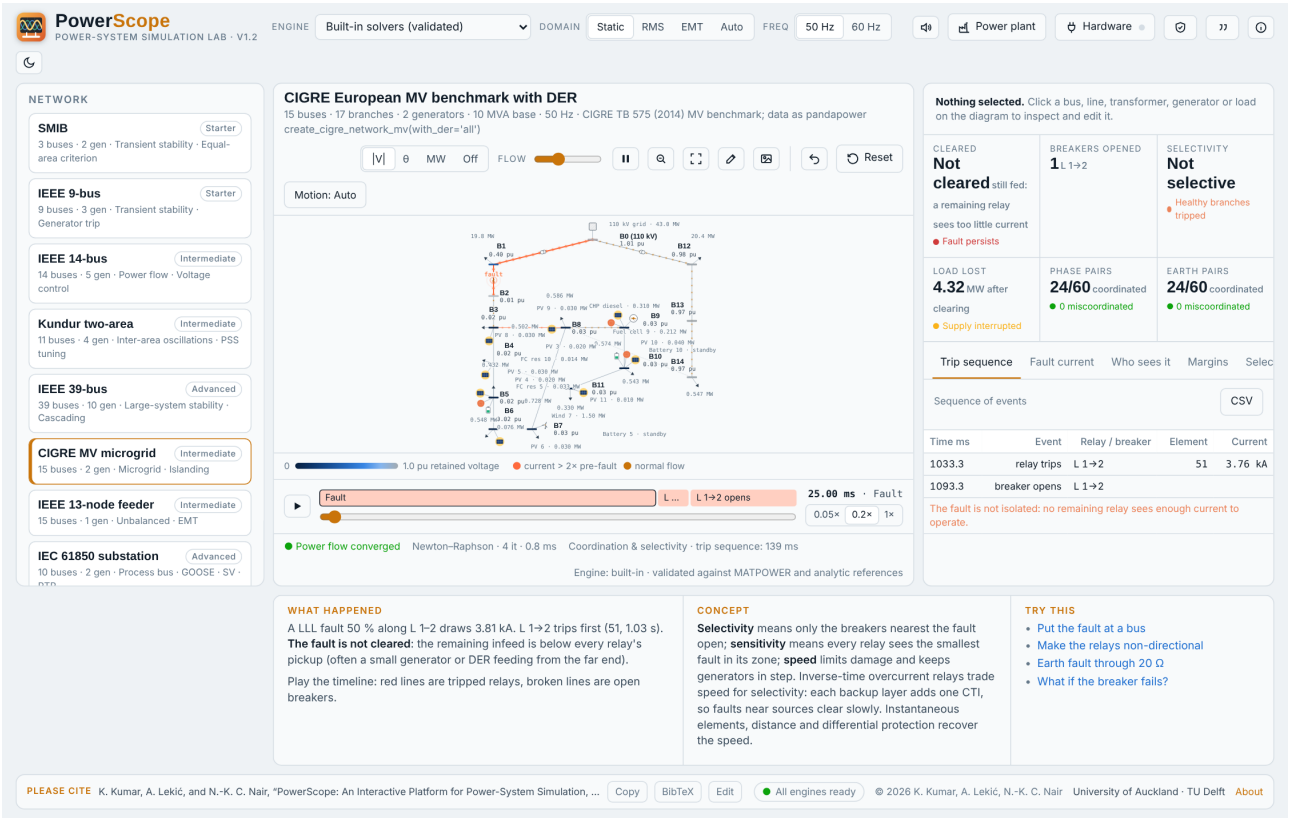


Figure 5. Protection: coordination and trip sequence on the CIGRE MV microgrid. In this default case the DER feeding from the far end keeps the fault alive after the feeder relay trips, so the study reports it as not cleared.

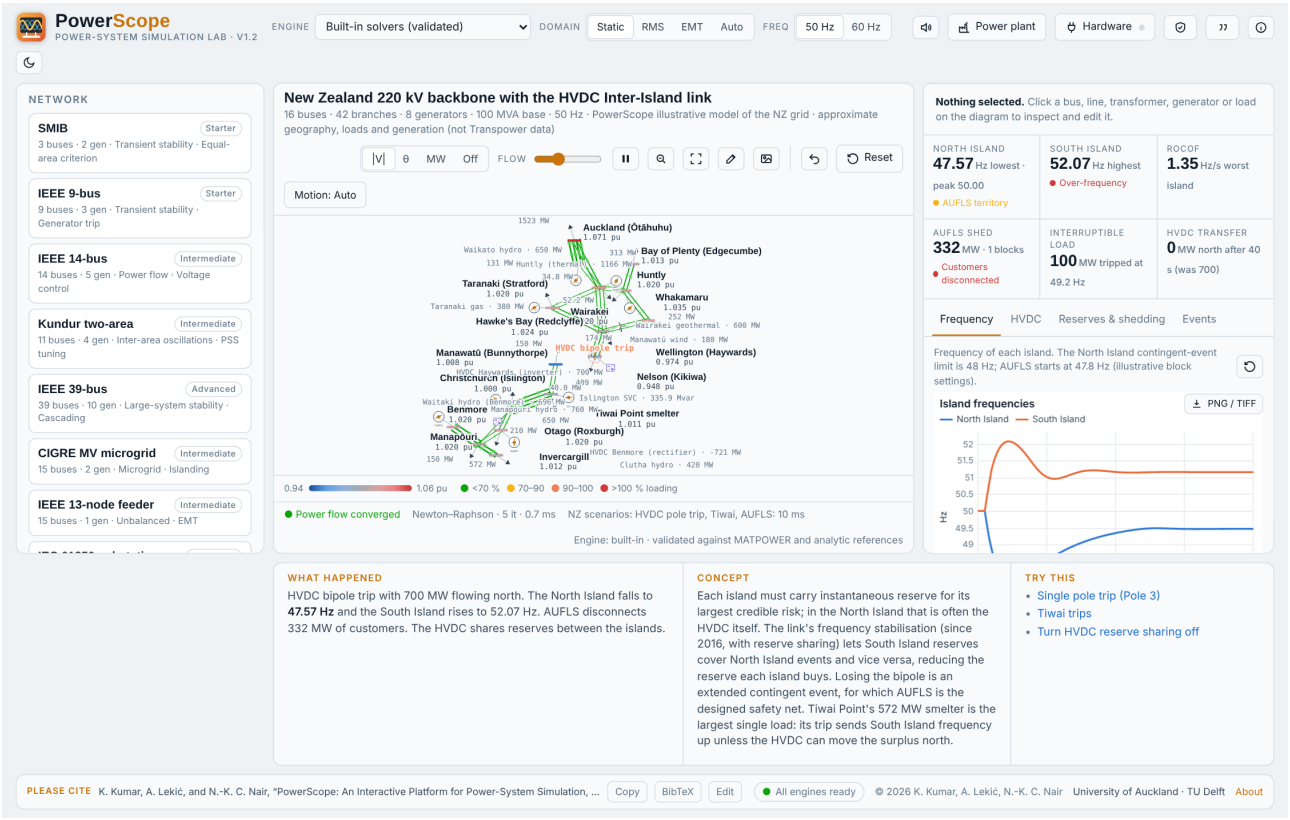


Figure 6. New Zealand pack: HVDC bipole trip with 700 MW flowing north, island frequencies and AUFLS.

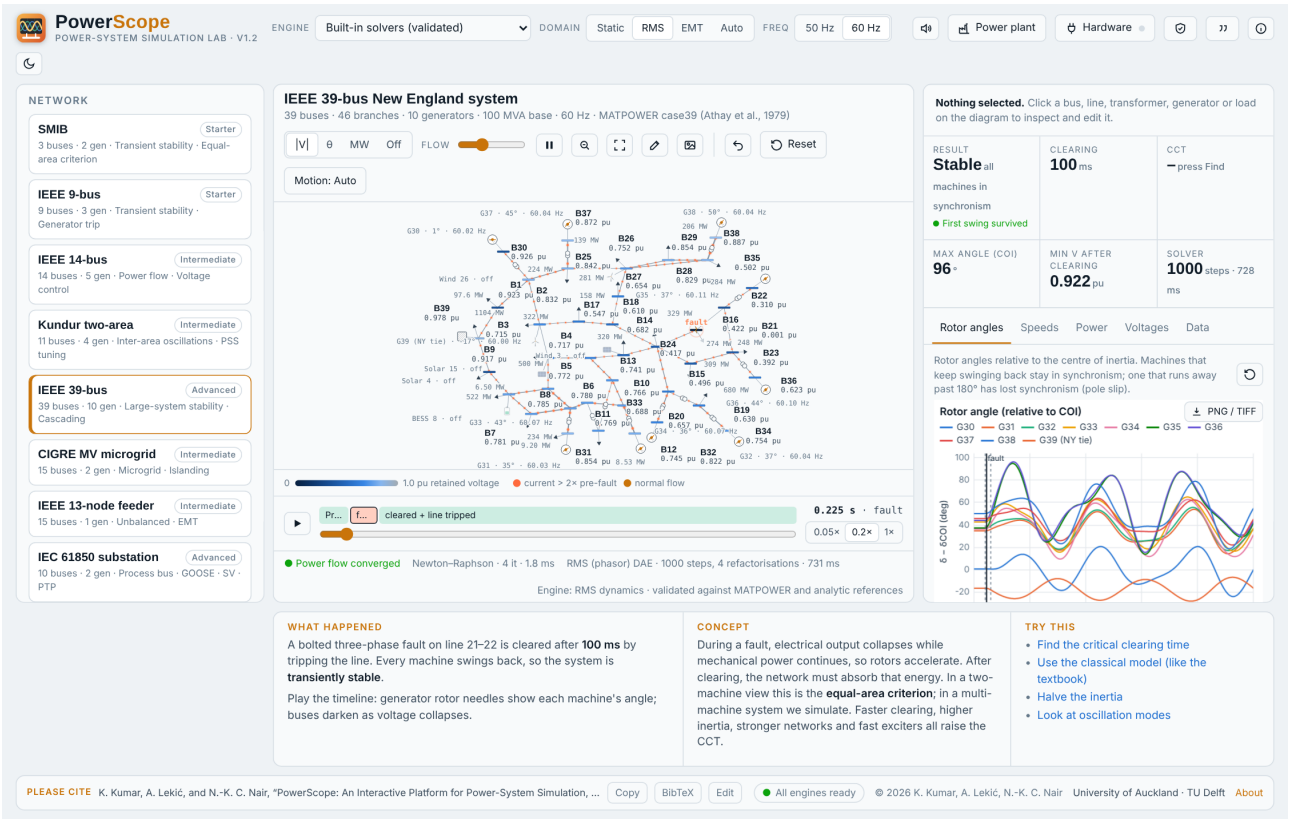


Figure 7. Transient stability on the IEEE 39-bus system.

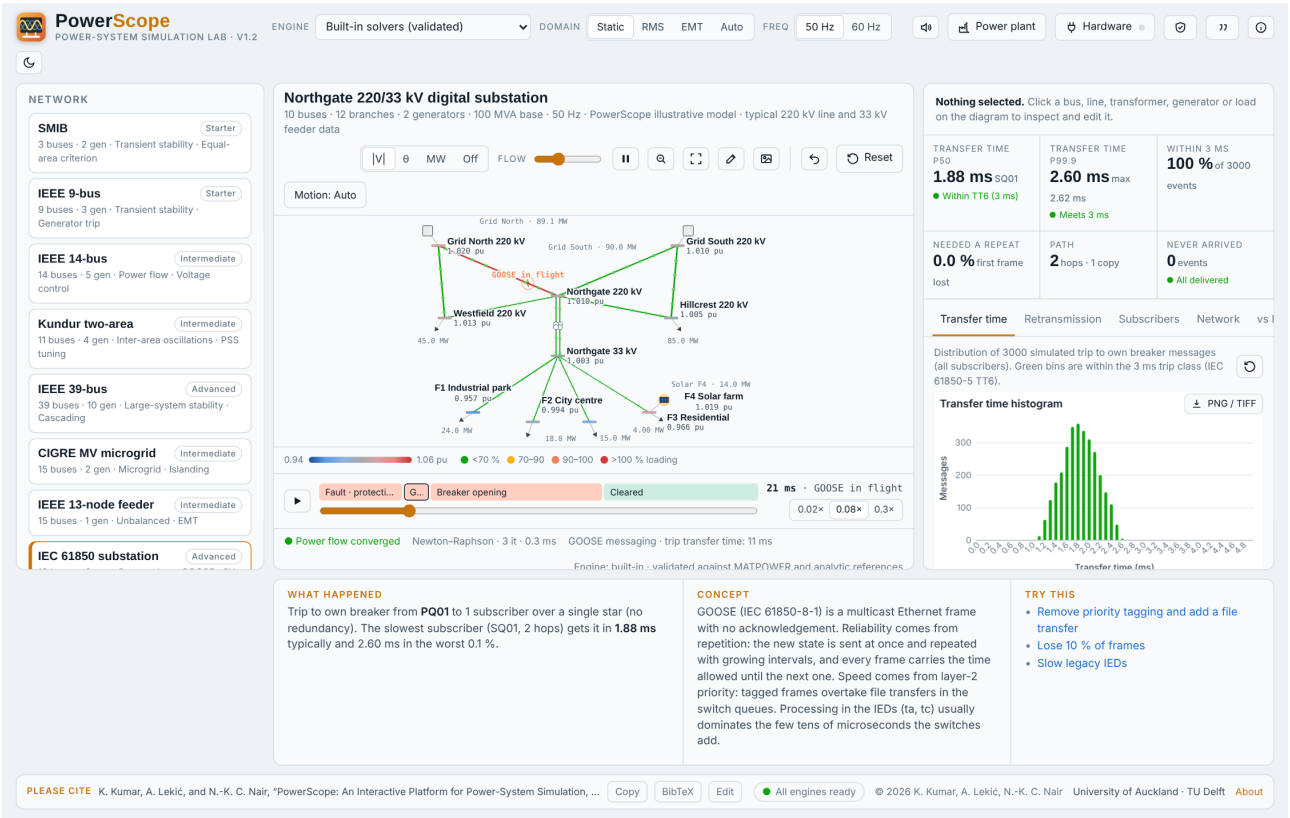


Figure 8. IEC 61850 digital substation: GOOSE messaging and trip transfer time.

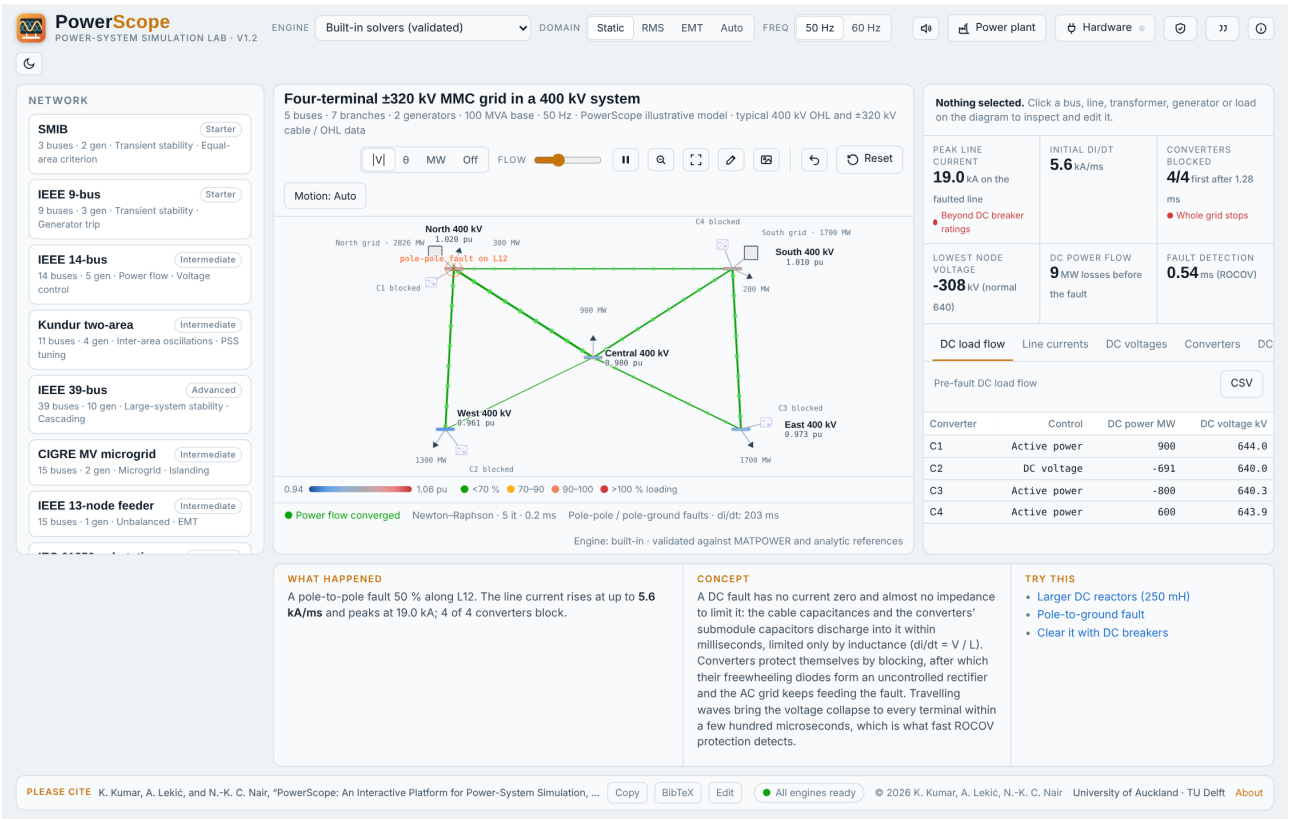


Figure 9. DC grid: pole-to-pole and pole-to-ground faults on the four-terminal MMC grid.

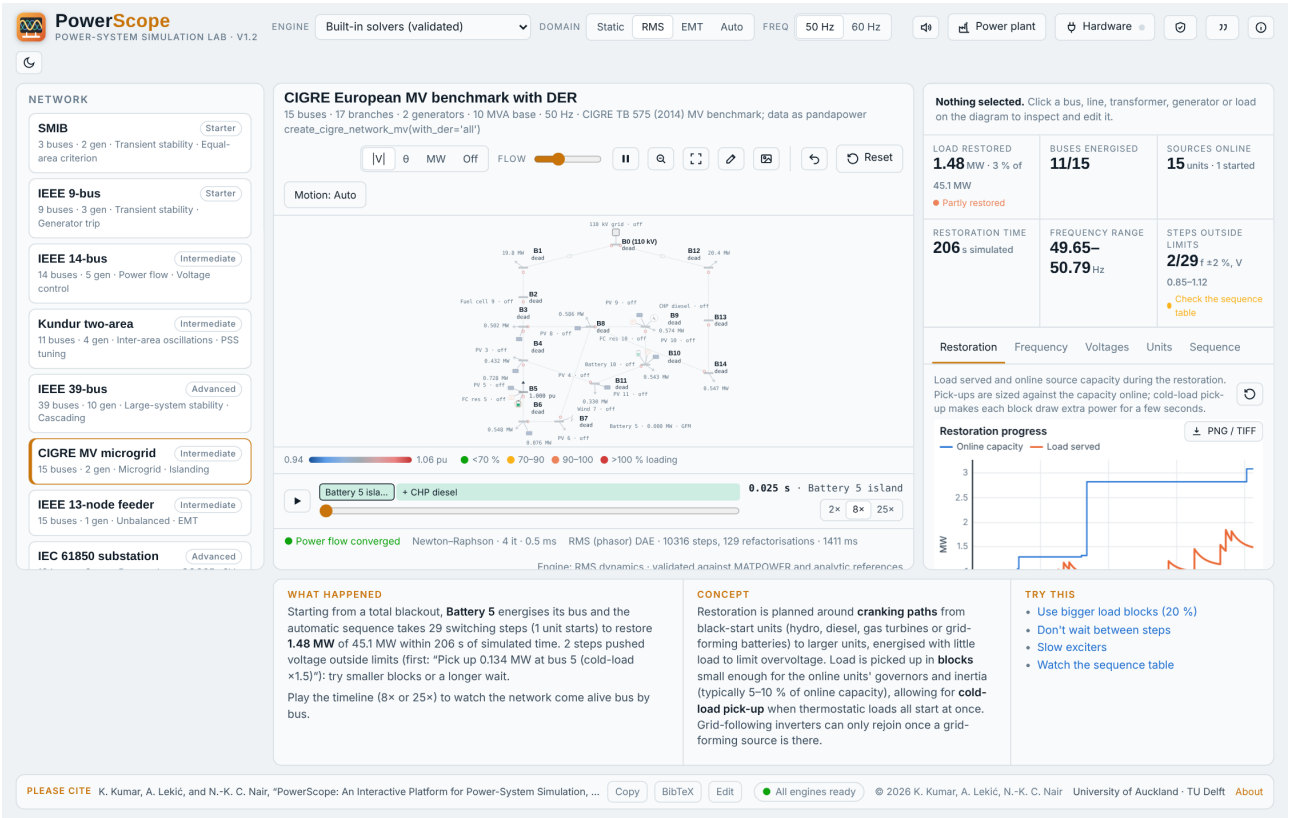


Figure 10. Black-start test and restoration path on the CIGRE MV microgrid.

12. Troubleshooting and limitations

Symptom	What to do
pandapower does not load	It needs an internet connection the first time and is blocked on some hosted pages. Use the offline copy of the file. The built-in engines are unaffected.
No Arduino port in the list	Use Chrome or Edge, close the Arduino IDE Serial Monitor (only one program can use the port), and install the CH340 driver for clone boards.
Cannot reach the Raspberry Pi bridge	A page served over https can only open ws://localhost or wss:// addresses; the offline copy can open any address.
Large EMT or restoration studies feel slow	These compute many thousands of steps; a busy indicator shows while they run. Smaller networks respond instantly.
Animation is distracting	Set Motion to Off, or let Auto follow the computer's reduce-motion setting.
Settings or citation edits disappeared	Preferences are kept in browser storage, which private windows and cleared site data remove.

Model limitations

- The New Zealand, IEC 61850 substation and HVDC networks are illustrative models with typical data, not utility data.
- Single-line studies on the IEEE 13-node feeder use a balanced equivalent; the unbalanced study uses the exact three-phase data.
- Results are for research and teaching. They do not replace detailed studies with utility data and certified tools.

13. Citation and licence

If PowerScope supports your research or teaching material, please cite it:

> K. Kumar, A. Lekić, and N.-K. C. Nair, “PowerScope: An Interactive Platform for Power-System Simulation, Protection Studies and Hardware-in-the-Loop Teaching,” under review, 2026.

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@misc{kumar2026powerscope,
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  title  = {{PowerScope: An Interactive Platform for Power-System Simulation,
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  note = {under review, 2026},
  year = {2026}
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Licence

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Benchmark data

IEEE and WSCC test systems (via MATPOWER), the CIGRE TB 575 MV benchmark, the Kundur two-area system and the IEEE 13-node feeder are used for verification. Institutional names identify the authors' affiliations; university logos are not used.